



## Original paper

# Toward the use of Silicon Carbide based detector for protontherapy microdosimetry

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## ABSTRACT

**Purpose:** This study explores the potential of a Silicon Carbide (SiC)-based microdosimeter for use in proton therapy. Thanks to its high radiation hardness, minimal leakage current, and stable charge collection efficiency, SiC is proposed as an innovative solid-state material for measuring microdosimetric quantities and enhancing radiobiological modeling in hadrontherapy.

**Methods:** A custom SiC p-n junction diode with a 10  $\mu\text{m}$  epitaxial layer was fabricated and electrically characterized. Microdosimetric spectra were acquired under proton beams at 30, 70, and 150 MeV at various depths in water. The SiC detector's performance was compared with benchmark microdosimeters, including a silicon-based MicroPlus probe and a single-crystal diamond device. Monte Carlo simulations were used to determine the water-equivalent thickness and to validate the measured spectra. Key microdosimetric quantities, such as frequency-mean and dose-mean lineal energy, were extracted and used to estimate the relative biological effectiveness (RBE) via the Microdosimetric Kinetic Model (MKM).

**Results:** The SiC detector demonstrated full charge collection efficiency and an energy resolution of approximately 2% under alpha irradiation. The measured spectra showed depth-dependent trends consistent with LET variations. The microdosimetric parameters derived from SiC data were in good agreement with both reference detectors and simulations. RBE values estimated from SiC measurements accurately reflected experimental data for U87 glioblastoma cells.

**Conclusion:** SiC-based detectors prove to be promising tools for microdosimetry in proton therapy. The consistency with reference systems and simulations supports their use in RBE estimation and biologically optimized treatment planning. Future work will target further miniaturization and application to heavier ions.

## 1. Introduction

Proton beam therapy is a well-consolidated approach in radiation oncology, offering superior dose conformity and improved sparing of healthy tissues compared to conventional photon-based radiotherapy [1,2]. Its key advantage lies in the characteristic Bragg peak, which enables precise energy deposition within the tumor while minimizing exposure to surrounding tissues. However, despite these physical benefits, significant radiobiological uncertainties remain, particularly regarding the spatial variation of relative biological effectiveness (RBE) along the beam path [3,4]. Current treatment planning assumes a fixed RBE of 1.1, disregarding variations in LET, dose fractionation, and tissue type. This oversimplification fails to account for the increased

LET near the distal Bragg peak, which may enhance biological damage, leading to potential underdosage within the tumor or unintended toxicity in adjacent critical structures [5,6].

To address these limitations, experimental microdosimetry has gained increasing attention as a tool for improving LET and RBE-based treatment planning systems (TPS). Microdosimetry provides a detailed characterization of the stochastic nature of energy deposition at a cellular scale, offering a more precise link between physical dose and biological damage. Unlike conventional dosimetry, which considers macroscopic dose distributions, microdosimetry focuses on the statistical fluctuations of energy imparted to microscopic volumes, enabling the study of radiation quality effects at subcellular dimensions [7,8]. In proton therapy, microdosimetric spectra can be

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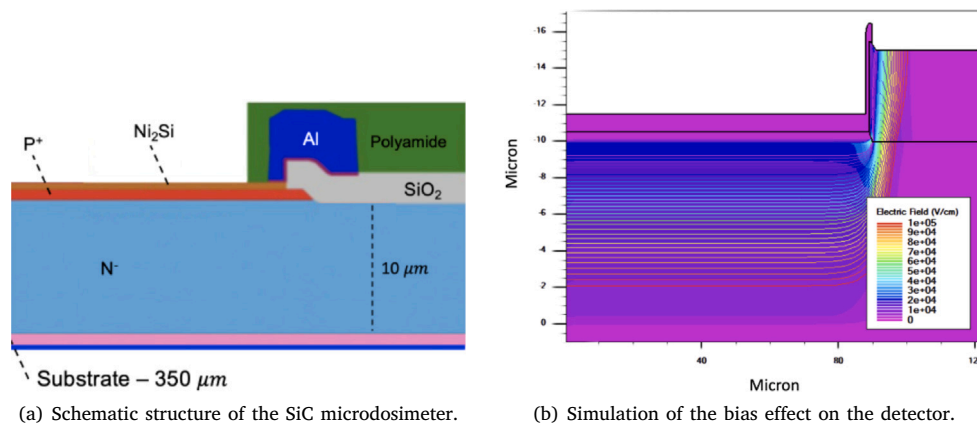
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**Fig. 1.** (a) Schematic representation of the SiC microdosimeter architecture, illustrating the epitaxial n-layer, p+ contact layer, and the overall detector layout. (b) Simulated electric field distribution inside the detector under a bias voltage of +10 V, highlighting the field concentration near the p–n junction and edge termination design.

used to assess variations in RBE as a function of LET and depth, supporting the development of more accurate biologically weighted dose models [9–11].

Over the past few decades, several types of microdosimetric detectors have been developed and tested for hadron therapy applications. Tissue-equivalent proportional counters (TEPCs) have historically been the reference detectors for microdosimetry, offering full tissue equivalence and accurate response to primary and secondary charged particles over a broad energy range. However, TEPCs suffer from several drawbacks, including pile-up effects, limited spatial resolution, and the need for high-voltage bias operation, which complicates their use in clinical settings [12]. To overcome these limitations, solid-state microdosimeters, particularly Silicon-on-Insulator (SOI) devices, have been developed as compact and high-resolution alternatives [10,13,14]. SOI-based microdosimeters feature micrometric sensitive volumes, allowing for fine spatial resolution, and are compatible with low-noise electronic readout systems. However, the major limitation of silicon-based detectors is their lack of tissue equivalence, which introduces energy-dependent calibration uncertainties and makes them susceptible to radiation-induced degradation, particularly at high doses [9].

An interesting and valuable alternative to the standard solid-state detectors is the so-called  $\Delta E - E$  telescope. This system consists of two detection stages with different thicknesses, typically referred to as  $\Delta E$  and  $E$ . The thinner  $\Delta E$  stage, usually a few micrometers thick, acts as a microdosimeter, enabling the measurement of microdosimetric spectra, while the thicker  $E$  stage measures the residual energy of the transmitted particles. This configuration not only provides microdosimetric information but also allows for ion species identification based on the combined  $\Delta E$  and  $E$  signals. Both silicon-based and diamond-based telescopes have been reported in the literature, offering excellent performance in terms of spatial resolution and energy discrimination. However, despite their advantages, the main technological limitation of these detectors lies in their minimum measurable LET value, which is constrained by the high electronic noise level associated with the detection system [15,16].

In recent years, diamond detectors have emerged as a promising alternative for microdosimetry in proton and heavy-ion therapy. Due to their near-tissue equivalence, high radiation tolerance, and excellent spectroscopic properties, single-crystal diamond detectors have been successfully employed in microdosimetric studies [17,18]. However, despite their advantages, diamond microdosimeters has a mean ionization energy of about 13 eV, higher than that of silicon (3.6 eV), which reduces the signals from detectors and, consequently, the signal-to-noise ratio. In addition, the fabrication technology is not as advanced as that of other semiconductors such as silicon. These constraints have

led researchers to explore alternative wide bandgap semiconductors, such as Silicon Carbide (SiC), which combines many of the desirable properties of silicon and diamond while mitigating their respective limitations [19].

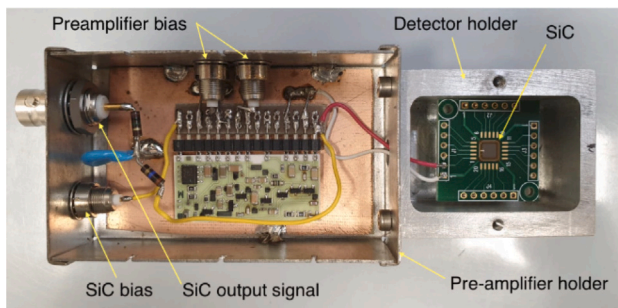
SiC has been extensively studied for radiation detection applications, particularly in high-energy physics and space environments, due to its intrinsic radiation hardness, low leakage current, and stable charge collection efficiency [20,21]. Compared to silicon, SiC exhibits a higher displacement energy (~25 eV vs. 13–20 eV for silicon), which reduces radiation-induced bulk damage and enhances detector longevity [22]. Additionally, the wide bandgap of SiC (3.26 eV for 4H-SiC) results in negligible dark current, ensuring low-noise operation even at room temperature (with an e-h energy of 7.78 eV). These characteristics make SiC an attractive candidate for high-resolution microdosimetry in particle therapy, particularly in environments with high radiation fluxes and long irradiation times [19,23].

To the best of our knowledge, this work represents the first experimental microdosimetric characterization of a therapeutic proton beam using a Silicon Carbide detector based on a p–n junction. The study aims to demonstrate the feasibility of SiC microdosimetry for both low and high energy proton beams (30 MeV, 70 MeV, and 150 MeV), providing a detailed assessment of lineal energy distributions, microdosimetric spectra, and dose-weighted microdosimetric quantities. The detector's performance was benchmarked against reference microdosimeters, including a silicon-based SOI device (MicroPlus probe) and a single-crystal diamond detector, to evaluate its response in different irradiation conditions. Additionally, Monte Carlo simulations were used to compare the experimental results with theoretical predictions, ensuring the accuracy of the measurements. The acquired microdosimetric spectra were further analyzed using the Microdosimetric Kinetic Model (MKM) to derive theoretical RBE predictions. These estimates were then benchmarked against radiobiological experimental data showing good agreement. The comparative analysis between modeled and experimental RBE values has offered valuable insights into the suitability of SiC microdosimetric approaches for proton therapy applications, highlighting the potential of this detector technology in enhancing treatment planning accuracy.

## 2. Material and methods

### 2.1. Device's design

The SiC microdosimeter used in this study consists of a p–n junction diode fabricated on a 4H-SiC substrate, a material well known for its superior high radiation tolerance. The detector sensitive volume



**Fig. 2.** Photograph of the assembled SiC detector mounted on a custom-designed Printed Circuit Board (PCB) and connected to the preamplifier inside its metallic enclosure.

is defined by an epitaxial n-layer with a thickness of 10  $\mu\text{m}$ , grown on a 350  $\mu\text{m}$ -thick SiC substrate. Over this, a  $\text{p}^+$  epi-layer (0.3  $\mu\text{m}$  to 1  $\mu\text{m}$  thick) is deposited to form the p–n junction. The active detection region is confined within the depletion layer of the diode, ensuring efficient charge collection and minimizing charge recombination effects. A schematic representation of the SiC microdosimeter layout is shown in Fig. 1(a). The detector has an active area of  $5 \times 5 \text{ mm}^2$ , resulting in a total active volume of 0.25  $\text{mm}^3$ .

The fabrication of the SiC microdosimeter has followed a multi-step photolithographic process, including high-temperature ion implantation, epitaxial growth, and metal contact deposition. The edge structure of the device was carefully designed to reduce electric field crowding, minimizing breakdown effects and ensuring uniform depletion under bias conditions. Fig. 1(b) presents the 2D electric field distribution inside the detector, simulated using the T-CAD simulator for a bias voltage of +10 V. The simulation highlights the electric field concentration near the p–n junction and at the detector edges, confirming the effectiveness of the edge termination design. The field distribution remains uniform across the active volume, ensuring optimal charge collection efficiency and stable detector performance.

The front contact is made of nickel, with additional aluminum and titanium layers, ensuring robust electrical connections and mechanical stability. After fabrication, the detector was mounted on a custom-designed Printed Circuit Board (PCB) for seamless integration with the readout electronics, as illustrated in Fig. 2.

The electronic chain adopted for microdosimetric spectra acquisition consists of an in-house designed preamplifier with a gain of 40 mV/MeV, physically connected to the detector case. The signal is processed by an Ortec 572 amplifier and digitized using a digital Ortec ADC-MCA (model EASY-MCA-8K), which converts the amplified pulses into 8192 channels. The ADC features a conversion time of approximately 2  $\mu\text{s}$  per event (including memory transfer), corresponding to a maximum throughput of several  $10^5$  cps. In our experiments, the actual rates were kept below  $10^3$  cps, ensuring negligible dead-time effect. The detector was negatively biased at –50 V using an Ortec 710 Quad Bias Supply. A picture of the assembled detector inside its metallic enclosure, connected to the preamplifier, is shown in Fig. 2.

## 2.2. Electrical and spectroscopic characterization

To assess the electrical properties and the suitability of the p–n junction diode, current–voltage (I–V) and capacitance–voltage (C–V) measurements were performed. The I–V characteristics, recorded at room temperature using a Keithley 6517B electrometer, were obtained by varying the applied voltage from –60 V to +2.5 V, with the Ni contact biased and the  $\text{p}^+$  layer grounded. As shown in Fig. 3(a), the device exhibited a rectification ratio of  $\sim 10^5$  at  $\pm 2$  V, indicating excellent diode performance. Furthermore, a low dark current, below 20 pA, was measured for a reverse bias up to –60 V. The inset in

Fig. 3(a) illustrates the forward region of the I–V curve, showing an exponential increase in current with applied voltage, as expected for a well-behaved p–n junction.

C–V measurements were conducted by applying voltage steps of 250 mV from –60 V to +2.5 V, using a Keithley 6517 electrometer, while capacitance was recorded with an Agilent 4284 A LCR meter operating at 1 kHz. The resulting C–V curve, shown in Fig. 3(b), confirms a fully depleted active region at approximately –30 V, corresponding to a depletion layer thickness of  $\sim 10 \mu\text{m}$ . The doping concentration of the epitaxial layer, estimated from the slope of  $1/C^2$  versus voltage, was found to be on the order of  $10^{14} \text{ cm}^{-3}$ , consistent with nominal values.

To evaluate the charge collection efficiency (CCE) and energy resolution of the SiC microdosimeter, a first test was performed using 5.5 MeV  $\alpha$ -particles from a  $^{241}\text{Am}$  source. The source and detector were positioned 3 cm apart inside a vacuum chamber to minimize air scattering, and a pinhole was used to collimate the particles. Given the penetration depth of  $\sim 18 \mu\text{m}$  for 5.5 MeV  $\alpha$ -particles in SiC, the detector was tilted at  $65^\circ$ , ensuring that the electron–hole pairs generated by the  $\alpha$ -particles were fully contained within the depletion region. The detector output was processed through a low-noise electronic chain, consisting of an Ortec 142 A charge preamplifier, a 3  $\mu\text{s}$  shaping amplifier (Ortec 570), and a multichannel analyzer (Ortec 927).

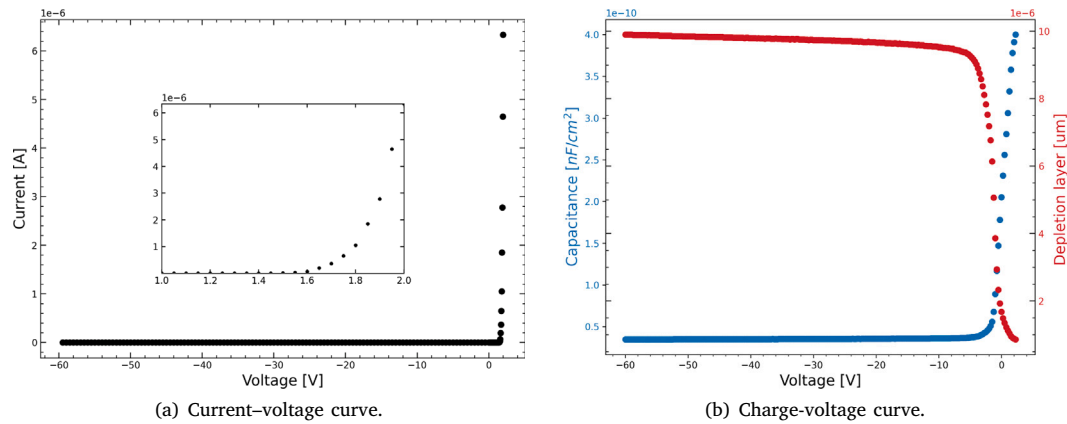
The electronic system was calibrated using a 100% charge collection efficiency silicon detector, and the charge collected in SiC was converted to energy assuming an average energy of 8.4 eV per electron–hole pair [20]. The pulse-height spectrum (PHS) for 5.5 MeV  $\alpha$ -particles at –50 V bias is presented in Fig. 4(a), demonstrating a well-defined peak with minimal broadening. A systematic study of energy resolution and CCE as a function of reverse bias was accordingly performed, with results summarized in Fig. 4(b). A saturation of both parameters was observed for bias voltages exceeding –40 V, with the SiC detector achieving an energy resolution of  $\sim 2\%$  and a CCE of 100%, confirming its excellent charge collection properties.

The energy calibration was performed by irradiating the detector in vacuum with monochromatic alpha particles from a natural  $^{148}\text{Gd}$  source, which emits alphas at 3.18 MeV. Lower energy alpha particles (2.50 MeV, 1.72 MeV, 1.22 MeV, and 0.46 MeV) were obtained by degrading the initial energy using aluminum absorbers of varying thicknesses, ensuring their complete absorption within the detector's active layer. The calibration response exhibited a linear dependence on the deposited energy, with an R-squared goodness-of-fit value of 0.987.

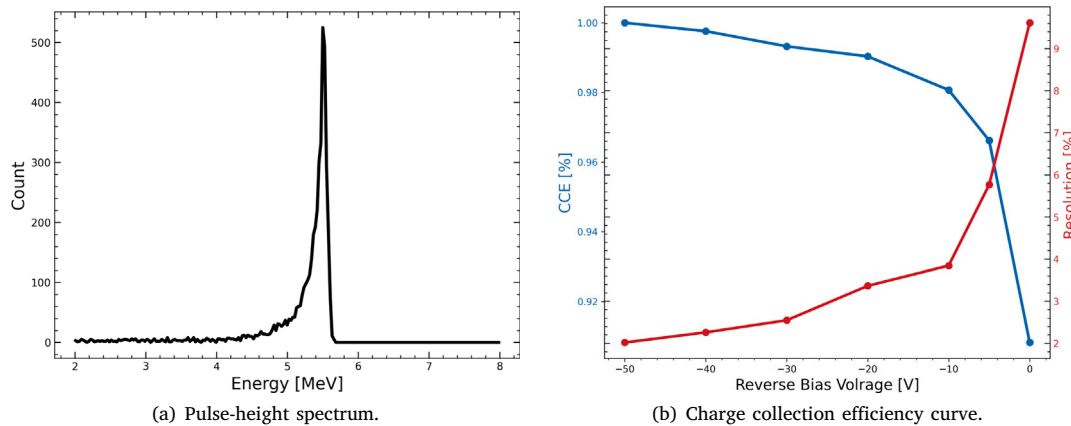
## 2.3. Experimental setup and irradiation conditions

To assess the performance of the SiC microdosimeter, three experimental campaigns were conducted using proton beams. In each experiment, microdosimetric spectra were acquired at different depths in water to evaluate the detector's response over a wide range of LET values. The obtained spectra were compared with reference measurements performed using two well-established detectors: the MicroPlus probe, based on silicon technology, and a single-crystal diamond detector [13,24]. Detailed descriptions of the two detectors are provided in the following sections. The irradiations were carried out using three different configurations referred to as A, B, and C:

- **Configuration A: 30 MeV Proton Beam** MicroPlus and SiC detectors were irradiated with monochromatic 30 MeV proton beams at the Nuclear Physics Institute of the Czech Academy of Sciences in Řež (Czech Republic). Due to the relatively low proton energy, this configuration was particularly useful for studying microdosimetric response at shallow penetration depths, where energy deposition is high and LET values are significantly greater compared to higher-energy beams.



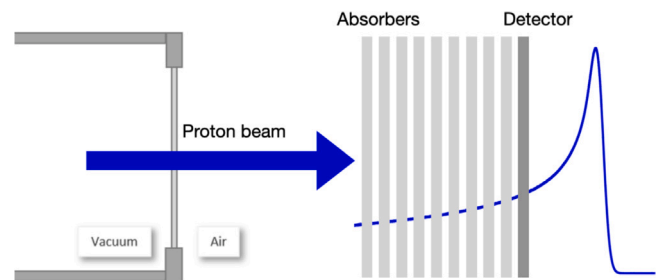
**Fig. 3.** (a) Current-voltage (I-V) characteristic curve of the SiC detector at room temperature, demonstrating its excellent rectification properties and low dark current. The inset shows a magnified view of the forward region of the curve. (b) capacitance-voltage (C-V) measurements performed to determine the depletion voltage and doping concentration of the epitaxial layer.



**Fig. 4.** (a) Pulse-height spectrum (PHS) obtained from the SiC microdosimeter under 5.5 MeV  $\alpha$ -particle irradiation at a reverse bias of  $-50$  V, demonstrating excellent energy resolution. (b) Charge collection efficiency (CCE) and energy resolution as a function of bias voltage, confirming 100% charge collection efficiency at voltages exceeding  $-40$  V.

- **Configuration B: 70 MeV Proton Beam** MicroPlus and SiC detectors were exposed to monochromatic 70 MeV proton beams at the Trento Proton Therapy Facility (Italy). This intermediate energy allowed for a more extended proton penetration in water, covering a broader LET spectrum and making it suitable for assessing the detector's response in conditions closer to those encountered in proton therapy applications.
- **Configuration C: 150 MeV Proton Beam** Diamond and SiC detectors were irradiated with monochromatic 150 MeV proton beams at the Trento facility. This high-energy configuration enabled measurements at deeper penetration depths and lower LET values, providing insights into the detector's response in conditions relevant to high-energy proton therapy.

At both irradiation facilities, the proton beams were extracted into air and directed toward a well-defined irradiation point where the detectors were placed. All accelerators used in this study are conventional isochronous cyclotrons. The RF frequencies are approximately 50 MHz at Řež and 106 MHz at Trento, corresponding to bunch widths of a few nanoseconds. Since these time structures are orders of magnitude shorter than the shaping times employed in our readout electronics (microseconds), the proton beams can be considered quasi-continuous for the present measurements. A schematic of the setup is shown in Fig. 5, depicting the proton beam passing through a series of absorbers



**Fig. 5.** Schematic representation of the experimental setup for proton irradiation. The figure illustrates the proton beam passing through absorbers and striking the microdosimeters positioned at different depths to acquire microdosimetric spectra.

positioned in front of the detectors. These absorbers, made of calibrated plastic foils, allowed for precise control of the measurement depth by adjusting the detector's position along the proton dose-deposition curve. Microdosimetric spectra were recorded at various depths for all configurations to analyze the detector response across different LET conditions.



To prevent signal pile-up, the proton beam current was reduced, ensuring a proton fluence on the detector in the range of  $10^4$  to  $10^7$   $\text{cm}^{-2}$ . The absence of pile-up was verified before each measurement by monitoring the detector output signal with a LeCroy WaveRunner 620Zi digital oscilloscope (2 GHz, 20 GS/s).

The depth-dose distribution was characterized before each experimental run using a stack of calibrated EBT-3 radiochromic films. The penetration range in water was defined as the depth at the 10% distal fall-off of the measured Bragg peak, resulting in values of 11.2 mm, 39.7 mm, and 156.4 mm for Configurations A, B, and C, respectively.

### 2.3.1. The MicroPlus probe detector

The MicroPlus probe, developed by the Centre for Medical Radiation Physics (CMRP) at the University of Wollongong, is a silicon-based microdosimeter designed for high-resolution measurements of energy deposition at the cellular scale. The detector consists of a three-dimensional array of cylindrical sensitive volumes (SVs), with dimensions comparable to those of biological cells. It is fabricated on a silicon-on-insulator wafer, where the active silicon layer has a thickness of  $10\text{ }\mu\text{m}$  [25] corresponding to a water-equivalent thickness of approximately  $30\text{ }\mu\text{m}$ .

The detector operates under a bias voltage of 9.5 V and is interfaced with a charge-sensitive electronic chain for signal processing. The output charge pulses, proportional to the energy deposited in each SV, are amplified using a charge-sensitive preamplifier and a shaping amplifier before being recorded by an ORTEC EASY-MCA multichannel analyzer connected to a computer for data acquisition. This system is based on the same EASY-MCA-8K digitizer used throughout the work, featuring a conversion time of approximately  $2\text{ }\mu\text{s}$  per event which corresponds to a maximum throughput of several  $10^5$  cps.

### 2.3.2. The diamond detector

The single-crystal diamond microdosimeter, developed at the Industrial Engineering Department of the University of Rome “Tor Vergata” features a multilayer structure, obtained through a two-step microwave plasma-enhanced chemical vapor deposition (MWPE-CVD) process on a synthetic diamond substrate. The microdosimeter operates as a Schottky diode, incorporating a boron-doped/intrinsic diamond/metal junction, fabricated using a combination of MWPE-CVD deposition and photolithographic techniques.

To precisely define the sensitive volume, a selective CVD diamond deposition method was used, resulting in a planar area of  $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$  and a thickness of  $12\text{ }\mu\text{m}$ , corresponding to a water-equivalent thickness of approximately  $38\text{ }\mu\text{m}$ . A 15 V bias voltage was applied during operation to optimize charge collection efficiency. Further details are available in [17,24]. The output signal was processed through a charge-sensitive preamplifier (Ortec 142) and a shaping amplifier (Ortec 570) before being recorded by an ORTEC EASY-MCA multichannel analyzer, which was interfaced with a computer for data acquisition. The entire electronic chain was calibrated for energy measurement.

## 2.4. Data analysis: Microdosimetric quantities and RBE prediction

The detectors used in this study can measure individual energy deposition events caused by incident particles within the active volume. This data was then processed into relevant microdosimetric quantities using the methodology outlined in [7]. The spectra obtained from the MCA were analyzed to extract key microdosimetric parameters essential for radiobiological modeling.

One of the fundamental quantities in microdosimetry, the lineal energy  $y$ , was extracted. This is defined as:

$$y = \frac{\epsilon}{l} \quad (1)$$

where  $\epsilon$  is the energy deposited in the sensitive volume (SV) during a single event, and  $l$  is the mean chord length within the SV. To enable a direct comparison between detectors, the water-equivalent mean chord

length was employed. By dividing the MCA-measured energy values by the corresponding water-equivalent mean chord length, the lineal energy  $y$  was obtained.

Microdosimetric spectra were then represented in terms of the lineal energy dose distribution,  $y_d(y)$ , where  $d(y)$  is the dose distribution function, mathematically expressed as:

$$d(y) = \frac{yf(y)}{y_F} \quad (2)$$

where  $f(y)$  is the lineal energy probability density function, and  $y_F$  is its expectation value, known as the frequency-mean lineal energy. This parameter is defined as:

$$y_F = \frac{\int yf(y)dy}{\int f(y)dy} \quad (3)$$

which represents the average lineal energy over all recorded events. The dose-mean lineal energy,  $y_D$ , corresponding to the second moment of the distribution, was calculated as:

$$y_D = \frac{\int y^2 f(y)dy}{\int yf(y)dy} \quad (4)$$

which provides a dose-weighted estimate of radiation quality. These parameters are essential for assessing the biological impact of proton radiation and were used to compare the performance of the SiC detector against reference microdosimeters.

To derive predictive estimates of the radiobiological effectiveness (RBE) of proton beams, the Microdosimetric Kinetic Model (MKM) [26] was employed using microdosimetric data from all of the adopted detectors. Specifically, the calculations were performed to derive the RBE vs. LET distributions at 10% clonogenic survival for the human glioblastoma cell line U87.

The MKM expresses the clonogenic survival probability through a modified linear-quadratic model (LQM), where the linear term incorporates the dose-mean lineal energy:

$$\alpha_{MKM} = \alpha_0 + \beta_0 \frac{y_D}{\rho \pi r_d^2} \quad (5)$$

Here,  $\alpha_0$  and  $\beta_0$  are cell-specific parameters that reflect the linear-quadratic model terms at the  $\text{LET} = 0$  limit,  $r_d$  is the mean radius of subnuclear domains which are critical for clonogenic survival, and  $\rho$  is the density of the cellular environment, equal to  $1\text{ g/cm}^3$ . By incorporating MKM within the LQM and fixing the MKM quadratic term to that of the reference radiation, the RBE for survival fraction  $S$  can then be calculated as:

$$\text{RBE}_S = \frac{\text{Dose}_{\text{ref}}}{\text{Dose}_{\text{exp}}} = \frac{\sqrt{\alpha_{\text{ref}}^2 - 4\beta_{\text{ref}} \ln S - \alpha_{\text{ref}}}}{\sqrt{\alpha_{\text{MKM}}^2 - 4\beta_{\text{ref}} \ln S - \alpha_{\text{MKM}}}} \quad (6)$$

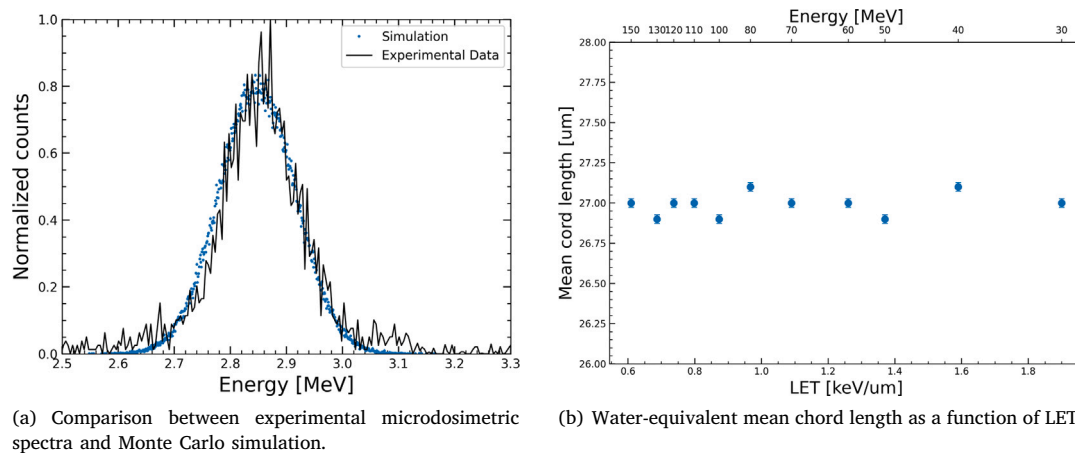
In the present work,  $\alpha_0$  and  $r_d$  were treated as free parameters and optimized to achieve the best agreement with RBE values from clonogenic survival curves conducted using the 62 MeV pristine proton beam at INFN-LNS [27,28]. The linear-quadratic parameters for the 225 kVp photon reference radiation,  $\alpha_{\text{ref}}$  and  $\beta_{\text{ref}}$ , were taken from an LQM fit performed on the same clonogenic curves [29], while  $\beta_0$  was set equal to that of the reference radiation. Specifically, for the U87 cell line, the linear-quadratic parameters for reference radiation exposure were determined as:

$$\alpha_{\text{ref}} = 0.14\text{ Gy}^{-1}, \quad \beta_{\text{ref}} = 0.051\text{ Gy}^{-2} \quad (7)$$

while model parameter optimization yielded:

$$\alpha_0 = 0.11\text{ Gy}^{-1}, \quad r_d = 0.3\text{ }\mu\text{m}. \quad (8)$$

An extension of the MKM, known as the modified-MKM (m-MKM) [30], incorporates corrections and saturation terms to account for the so-called overkill-effect, which becomes relevant at very high lineal energy values. However, since protons are a radiation type with a relatively low LET, applying the m-MKM in the present work would lead to negligible differences compared to the standard MKM.



**Fig. 6.** (a) Comparison between experimental microdosimetric spectra acquired with the SiC detector and Monte Carlo simulated spectra at different depths, validating the detector response against theoretical predictions. (b) Relationship between the SiC detector's water-equivalent mean chord length and the corresponding proton energy and LET values, as determined from Monte Carlo simulations.

### 2.5. Monte Carlo simulations

Monte Carlo simulations using the Geant4 toolkit were conducted to determine the SiC detector's equivalent water thickness and to verify that the mean values derived from the microdosimetric spectra aligned with the expected LET values [31]. The equivalent thickness, as used here, does not correspond to the physical slab thickness of the detector but to the *effective water-equivalent mean chord length*, determined by Geant4 simulations matching the SiC response to an equivalent water volume. This effective mean chord length is the reference quantity employed for lineal energy normalization throughout this work. The equivalent thickness was estimated using a dedicated Geant4 application following the methodology described by Guatelli et al. [32]. To assess the water equivalency of SiC at different depths within a phantom, a computational model was developed to study the energy deposition process at the microscopic level, taking into account the interactions of both primary and secondary particles. Hadronic interactions, including both elastic and inelastic processes, were simulated using the G4HadronElasticPhysicsHP, G4IonElasticPhysics, G4HadronPhysicsQGSP\_BIC\_HP, and G4IonPhysicsSPHP models. Electromagnetic interactions were modeled using the G4EmStandardPhysics\_option4 package.

A detailed simulation of the detector geometry was implemented, including the inactive SiC substrate and the metallization dead layers preceding the active detection region. To validate the detector model, the experimental setup used for characterization was replicated in the simulation, and the resulting spectra were compared with experimental data (see Fig. 6(a)).

The mean chord length in water was determined through an iterative simulation process. First, the energy spectrum of the proton beam was simulated within the active SiC layer. A second simulation was then performed under identical irradiation conditions, replacing the SiC active layer with water. The water-equivalent thickness was gradually adjusted until the simulated spectra for SiC and water matched perfectly. The water thickness at which this agreement was reached was adopted as the mean chord length in water. This value was subsequently used for reconstructing the experimental microdosimetric spectra.

The relationship between the SiC mean chord length in water, LET, and incident kinetic energy is presented in Fig. 6(b).

To reconstruct the expected LET values for each irradiation configuration, the *Radiobiology* extended example from the official Geant4 release was employed. This open-source application is specifically designed for dosimetric and radiobiological studies involving proton and

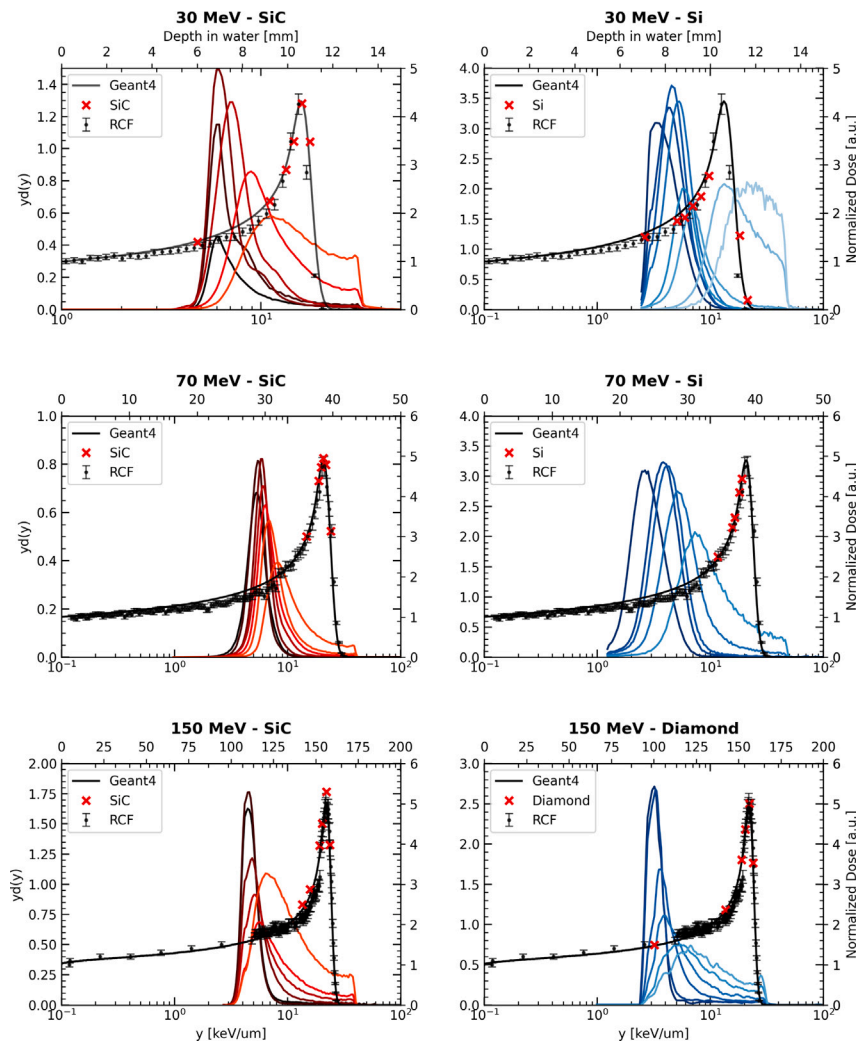
ion beams, incorporating validated algorithms for computing key radiobiological parameters such as Dose, LET, and RBE. To accurately replicate the experimental irradiation conditions for Configurations A, B, and C, the source term was modified accordingly. The proton beam was modeled with a circular spot shape, assuming a bivariate Gaussian distribution with a standard deviation of 2 mm. Additionally, an angular distribution with a standard deviation of  $0.028^\circ$  was implemented. For Configuration A, the primary proton beam was simulated with a Gaussian energy distribution centered at 34.3 MeV and a standard deviation of 0.5 MeV, as provided by the facility staff at the Nuclear Physics Institute in Řež (no published reference available). For Configurations B and C, the beams were centered at 68.8 MeV and 150 MeV, respectively, each with a standard deviation of 0.6 MeV, consistent with the characterization of the Trento Proton Therapy Centre experimental beam line [33]. In all cases, a Gaussian angular spread with  $\sigma = 0.028^\circ$  was included to account for beam divergence. All simulations were performed using Geant4 version 10.08.p02 (May 2024), with each run consisting of  $1 \times 10^7$  primary events [31].

## 3. Results and discussion

### 3.1. Microdosimetric spectra and comparison with Monte Carlo simulations

The microdosimetric dose distributions obtained with the SiC detector, alongside the reference detectors (Microplus probe and diamond), are shown in Fig. 7. The data are presented as  $yd(y)$  versus  $y$  on a logarithmic scale to improve visualization and interpretation. In this representation, the area under the curve between two  $y$ -values quantifies the fraction of the absorbed dose associated with energy deposition events within the corresponding  $y$ -interval [7].

The spectra measured with the SiC detector exhibit a cut-off around  $3 \text{ keV } \mu\text{m}^{-1}$ , corresponding to the detection system's electronic noise threshold during irradiation. The low signal-to-noise ratio at the entrance of the Bragg peak, where the LET is low and the incident proton energy is high, prevented the acquisition of reliable microdosimetric spectra. However, a consistent trend emerges with increasing depths, as shown in Fig. 7: the dark red curve, positioned furthest to the left, represents the proximal region, while the light red curves on the right correspond to measurements taken beyond the Bragg peak. As protons penetrate deeper into the medium, the distributions progressively shift towards higher  $y$ -values, reflecting the increase in LET. This behavior culminates near the Bragg peak, where the ionization density reaches its maximum as the protons come to rest.



**Fig. 7.** Microdosimetric spectra acquired at various depths in water with the SiC detector (red curves) and reference detectors (MicroPlus silicon probe and diamond microdosimeter, blue curves). The shift of the spectra toward higher lineal energy values with increasing depth reflects the expected LET increase near the Bragg peak.

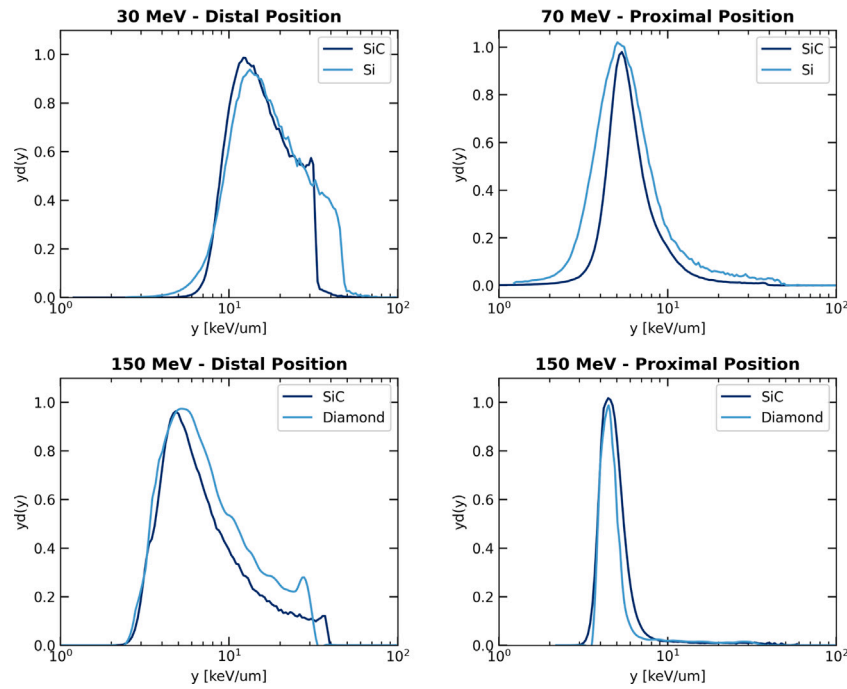
This trend aligns perfectly with expectations and is consistently observed for both reference detectors. Specifically, the entrance region is characterized by lower  $y$ -values, represented by the dark blue curves, whereas distal regions, beyond the Bragg peak, exhibit higher  $y$ -values, depicted by light blue curves (see Fig. 7, right side).

A small enhancement at the proton edge was observed in the spectra acquired with the SiC detector at 30 MeV. This feature is consistent with a moderate pile-up effect. The relatively large sensitive area of the detector increases the probability of multiple protons being detected within the shaping time of the readout electronics. In addition, to guarantee a fluence of  $\sim 10^3$  protons at the detector after the absorbers, the beam current at the entrance was increased; however, the actual count rate could not always be precisely controlled and was likely higher than expected, thereby favoring pile-up. The observed effect is thus compatible with count rates in the  $10^2$ – $10^3$  cps range, occasionally exceeding this value, in agreement with qualitative oscilloscope checks during the measurement. Furthermore, in all acquired spectra, both with the SiC detector and the reference detectors, the so-called proton edge ( $y_{\text{proton}}$ ) is clearly visible. This feature corresponds to the rightmost part of the spectra, where the distributions sharply drop to zero. It represents the maximum energy that protons can deposit within the SV. The silicon-based detector, the MicroPlus probe, with an active thickness of  $10\text{ }\mu\text{m}$ , allows protons to deposit a maximum energy of approximately  $736\text{ keV}$ , which corresponds to a lineal energy  $y_{\text{proton}}$

in water of approximately  $43\text{ keV }\mu\text{m}^{-1}$ . In contrast, for the diamond-based microdosimeter, with an active thickness of  $12\text{ }\mu\text{m}$ , the maximum energy deposition reaches approximately  $1300\text{ keV}$ , yielding a  $y_{\text{proton}}$  of about  $34\text{ keV }\mu\text{m}^{-1}$  in water. Finally, the SiC detector, with a maximum energy deposition of approximately  $848\text{ keV}$ , exhibits a proton edge corresponding to  $38\text{ keV }\mu\text{m}^{-1}$  in water. These discrepancies account for the small observed differences between the microdosimetric distributions obtained with the different detectors, reflecting the distinct energy deposition capacities of each device based on their SV thickness and material properties.

A deeper insight into the microdosimetric spectra acquired with the SiC detector was achieved by considering both the SV's geometrical and dimensional characteristics. A comparison between the proximal and distal regions of the spectra was conducted, using the reference detectors as benchmarks. Fig. 8 shows the lineal energy distributions measured with the SiC detector (dark blue lines) against those obtained with the reference (light blue lines).

The comparison reveals that the microdosimetric spectrum recorded with the silicon-based Microplus probe is markedly broader than that obtained with the SiC detector, exhibiting pronounced tails at both low and high LET values. This broadening effect stems from the detector's geometry: the Microplus probe, characterized by cylindrical SVs, more accurately reproduces the isotropic nature of radiation tracks, in contrast to the planar architecture of the SiC detector, which is inherently more sensitive to the angular distribution of the incident particles.



**Fig. 8.** Comparison of microdosimetric spectra obtained with the SiC detector and reference detectors (MicroPlus probe and diamond microdosimeter), demonstrating the influence of detector geometry on energy deposition distributions.

Conversely, the spectra acquired with both the diamond and SiC detectors are comparable in shape and in the position of the primary peak, as expected due to their shared slab geometry. Minor differences are observed in the location of the proton-edge and in the width of the main peak. Notably, the proton-edge appears at a slightly lower energy for the diamond detector, which features a marginally thicker sensitive volume compared to SiC. This behavior, observed in water-equivalent terms, highlights the influence of detector thickness on the energy deposition profile. The proton-edge position effectively extends the microdosimetric spectrum of thinner detectors toward higher lineal energy. Furthermore, the spectrum of SiC, being thinner than that of diamond, appears broader due to the increased energy-loss straggling, which results from the shorter particle path length within the detector.

Finally, to assess the performance of the SiC detector in reconstructing the LET distribution as a function of depth, the experimental  $y_D$  and  $y_F$  were calculated. These quantities represent the dose-mean and frequency-mean lineal energy, respectively, and were derived from the experimental microdosimetric spectra following the methodology described in the previous section. The resulting values were subsequently compared with the average LET values obtained through Monte Carlo simulations, in accordance with the approach outlined by Cortés-Giraldo and Carabe [34] and Conte et al. [12], further implemented and validated by Petringa et al. [35].

Fig. 9 presents the simulated  $L_{\text{track}}^{\text{total}}$  and  $L_{\text{dose}}^{\text{total}}$  alongside the corresponding experimental  $y_D$  and  $y_F$  values derived from the acquired data. The  $L_{\text{total}}$  is determined by considering not only the contribution from the average LET of the interacting primary protons but also accounting for the secondary particles produced during the interaction, along with their respective LET values. These secondary particles, which include ions and other fragments, also contribute to the overall energy deposition in the medium, thereby playing a significant role in the total LET calculation. The inclusion of these secondary contributions is essential for accurately modeling the energy release and the biological effects of the radiation. The  $y_D$  and  $y_F$  values calculated with the SiC detector show excellent agreement with those obtained using silicon and diamond microdosimeters, as well as with the simulated total LET. This consistency underscores the reliability of the SiC device in microdosimetric applications and confirms its sensitivity to the

contribution of secondary particles generated by nuclear interactions within the absorber layers [12,35].

The dose-mean lineal energy  $y_D$  measured with the SiC microdosimeter exhibits an increase up to approximately  $10 \text{ keV } \mu\text{m}^{-1}$  in the Bragg peak region, reaching a maximum of around  $20 \text{ keV } \mu\text{m}^{-1}$  at the distal fall-off region. The  $y_D$  values measured across the entire water depth are in good agreement with those obtained with the reference detectors, confirming the detector's reliability in reproducing the depth-dose profile.

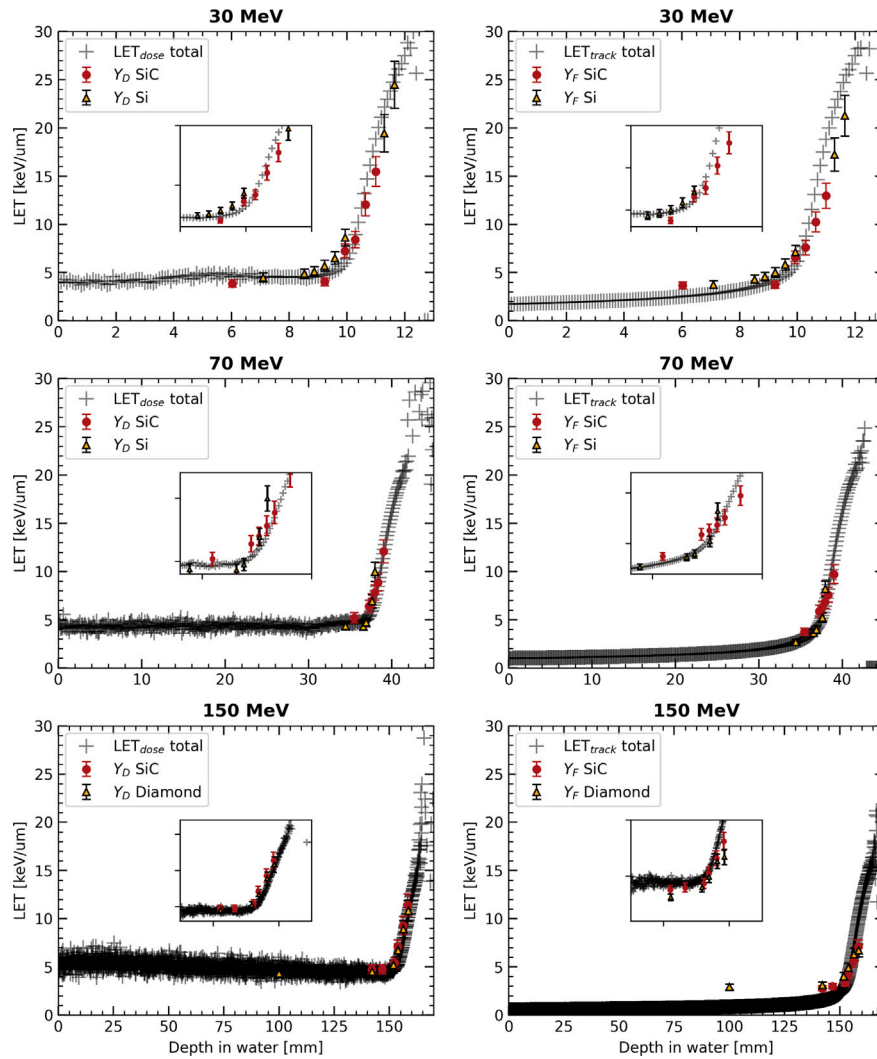
Conversely, the frequency-mean lineal energy  $y_F$  measured by the SiC detector remains consistent with the reference detectors throughout the entire water depth. Specifically,  $y_F$  values amount to approximately  $1 \text{ keV } \mu\text{m}^{-1}$  at the entrance region, progressively increasing to a maximum of about  $15 \text{ keV } \mu\text{m}^{-1}$  at the distal fall-off region, in line with expectations based on the radiation field characteristics.

A chi-square goodness-of-fit test was performed to quantitatively evaluate the agreement between the experimental lineal energy distributions and the expected LET values. The chi-square test was performed with a confidence level of 95% ( $\alpha = 0.05$ ) to assess the goodness of fit between the experimental microdosimetric quantities acquired with the SiC detector and the expected LET values simulated through Geant4. The chi-square values and p-values were calculated for each energy configuration, considering both the total and primary LET components as reported in Table 1.

The results show that at all energies, the p-values are consistently close to 1, indicating that there is no statistically significant difference between the two sets of data. However, the best agreement is observed at 150 MeV, where the chi-square values are lower across all comparisons, especially for  $y_D$  versus  $L_{\text{dose}}^{\text{total}}$  ( $\chi^2 = 0.0035$ ,  $p = 1.0000$ ) and  $y_D$  versus  $L_{\text{dose}}$  ( $\chi^2 = 0.1671$ ,  $p = 0.9994$ ). This suggests that the microdosimetric quantities at higher proton energies are more consistent with the expected LET values, likely due to the more uniform energy deposition and reduced statistical fluctuations at higher energies.

The largest chi-square values were observed at 30 MeV, particularly for the dose-mean quantities, indicating that at lower proton energies, the microdosimetric distributions are more affected by straggling effects and stochastic fluctuations. The intermediate energy of 70 MeV





**Fig. 9.** Depth dependence of the dose-mean lineal energy ( $y_D$ ) and frequency-mean lineal energy ( $y_F$ ) derived from SiC microdosimetric spectra, compared with total LET values obtained from Monte Carlo simulations.

**Table 1**

Chi-square and p-values obtained from the statistical comparison between experimental microdosimetric quantities ( $y_D$  and  $y_F$ ) measured with the SiC detector and the expected LET values derived from Monte Carlo simulations. The comparisons are provided for different proton energies (30 MeV, 70 MeV, and 150 MeV), considering both total and primary LET components.

|         |          | $y_D/L_{dose}^{total}$ | $y_D/L_{dose}$ | $y_F/L_{track}^{total}$ | $y_F/L_{track}$ |
|---------|----------|------------------------|----------------|-------------------------|-----------------|
| 30 MeV  | $\chi^2$ | 0.081                  | 0.340          | 0.180                   | 0.190           |
|         | p-value  | 0.999                  | 0.997          | 0.999                   | 0.999           |
| 70 MeV  | $\chi^2$ | 0.395                  | 0.641          | 0.120                   | 0.121           |
|         | p-value  | 0.995                  | 0.985          | 0.998                   | 0.998           |
| 150 MeV | $\chi^2$ | 0.003                  | 0.167          | 0.080                   | 0.087           |
|         | p-value  | 1.000                  | 0.999          | 0.999                   | 0.999           |

shows an overall improvement compared to 30 MeV but still exhibits larger chi-square values compared to 150 MeV. These findings highlight the energy dependence of microdosimetric measurements and confirm that higher proton energies yield more stable and predictable microdosimetric distributions.

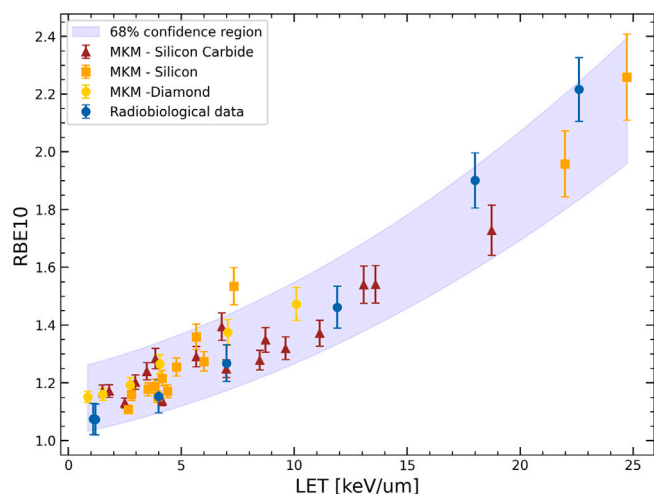
### 3.2. Potential implications for RBE predictions

The capabilities of the SiC-based microdosimeter in predicting biological damage were evaluated by applying the MKM as described

in Section 2.4. The same procedure was used to estimate the RBE with the reference detectors (MicroPlus probe and diamond) across all studied energy configurations. RBE values corresponding to 10% clonogenic survival ( $RBE_{10}$ ) were derived for the U87 cell line. The selection of the U87 cell line was motivated by its widespread use in cancer research, particularly in glioblastoma studies, due to its well-characterized radioresistance and clinical relevance. This makes U87 a robust model for assessing the radiobiological impact of proton radiation [36,37].

For the validation of the model's predictions, experimental data reported in [27] were employed. These data were obtained by positioning biological samples at six distinct locations along the monoenergetic 62 MeV proton Bragg peak, allowing for the investigation of the variation in cellular response as a function of LET. The biological endpoint was assessed using the clonogenic assay, which provides a quantitative measure of cell survival following irradiation. Fig. 10 displays the  $RBE_{10}$  values obtained from the MKM, illustrating the RBE trends as a function of the LET. The model's predictions are compared with experimental data from in-vitro clonogenic survival assays. To account for the experimental uncertainties, the figure includes a shaded blue region representing the 68% confidence interval ( $1\sigma$  uncertainty). This confidence interval encompasses the measurement errors taking into account the intrinsic variability of the experimental data.

Remarkably, all  $RBE_{10}$  values predicted by the MKM fall within this confidence region, underscoring the model's robustness in reproducing



**Fig. 10.** RBE as a function of LET for proton beams of different energies. The RBE predictions obtained using the Microdosimetric Kinetic Model with SiC microdosimetric data, Microplus probe and Diamond are compared with experimental clonogenic survival data.

the experimental observations across different beam configurations and adopting different detectors. Among the three detectors employed, the silicon carbide-based detector demonstrated well-centered predictions over a wide range of LET values. This highlights its capability in accurately measuring the key physical mechanisms underlying RBE, including energy deposition patterns and the track structure of ionizing radiation.

Consequently, the RBE estimation derived from the SiC detector exhibits high accuracy, effectively reflecting the biological effects of proton radiation across varying LET values. By providing reliable RBE estimations, SiC detectors offer valuable insights for optimizing radiotherapy treatment protocols, especially in hadrontherapy where the RBE varies significantly with particle energy and LET.

#### 4. Conclusion

The present study represents the first comprehensive experimental investigation of a SiC-based detector for microdosimetric applications in proton therapy, confirming the feasibility of this emerging technology for the precise characterization of lineal energy distributions and dose-weighted microdosimetric quantities.

The detector was first electrically characterized to determine the active thickness of the sensitive volume, a key parameter for calculating the mean chord length in microdosimetry. Spectroscopic tests followed, revealing a full charge collection efficiency and an energy resolution of approximately 2%, thereby validating the detector's high signal integrity and suitability for accurate energy-loss measurements. Monte Carlo simulations were also employed to estimate the detector's response, with particular attention to the mean-chord length estimation in water.

Subsequently, the SiC detector was exposed to proton beams of 30 MeV, 70 MeV, and 150 MeV, enabling a thorough assessment of its microdosimetric performance across a wide range of LET conditions. The detector showed excellent agreement with reference microdosimetric devices, such as the silicon-based MicroPlus probe and single-crystal diamond detectors, reliably reconstructing key quantities like the dose-mean lineal energy ( $y_D$ ) and frequency-mean lineal energy ( $y_F$ ). The acquired microdosimetric spectra demonstrated a consistent shift towards higher  $y$ -values with increasing depths along the Bragg curve, in line with the expected rise in LET. This trend was further corroborated by Monte Carlo simulations, which showed consistency between the

measured microdosimetric quantities and the simulated LET distributions, with chi-square goodness-of-fit tests yielding p-values close to unity across all investigated proton energies. The largest discrepancies were observed at low proton energies (30 MeV), where statistical fluctuations and straggling effects play a more prominent role, whereas the best agreement was found at higher energies (150 MeV), where the microdosimetric distributions become more stable and predictable.

In addition to its excellent agreement with theoretical predictions, the SiC detector demonstrated high charge collection efficiency, low electronic noise, and robust radiation tolerance, making it a promising candidate for high-flux irradiation environments. The application of the MKM to the experimental data further highlighted the detector's potential for radiobiological modeling, with  $RBE_{10}$  predictions for the U87 glioblastoma cell line showing good agreement with in-vitro experimental data. Notably, the SiC detector exhibited the most accurate RBE predictions among the three detectors, underscoring its suitability for enhancing biologically weighted treatment planning systems in proton therapy.

While these results are promising, several aspects of the detector design could still be optimized to enhance its microdosimetric performance. The planar geometry of the SiC detector is advantageous for fabrication and integration. However, it introduces angular dependencies that may limit its ability to reproduce the isotropic radiation track structure. Future developments could focus on fabricating micrometric sensitive volumes with reduced dimension and with three-dimensional geometries, similar to those used in silicon-based microdosimeters. This approach could enhance the detector's spatial resolution and angular response. Additionally, a limitation of the present work is that only quasi-continuous beams from isochronous cyclotrons were investigated. In synchro-cyclotrons, operating at pulse frequencies in the kHz range, the instantaneous particle flux is significantly higher, which may lead to increased pile-up with the current readout chain. Nonetheless, with proper adjustment of beam intensity and by employing faster digitizers, the SiC microdosimeter could be adapted to synchro-cyclotron facilities. This represents an important direction for future developments. In conclusion, the results presented in this work demonstrate that SiC-based detectors are a viable alternative to conventional microdosimetric technologies, offering a unique combination of radiation hardness, and accurate microdosimetric response. The agreement between experimental and theoretical microdosimetric quantities, coupled with the detector's robust RBE predictions, highlights its potential for advancing the field of protontherapy microdosimetry. Further research and technological advancements will be crucial to fully harness the capabilities of SiC detectors, paving the way for their integration into future treatment planning systems for precision radiotherapy applications.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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